TEK306- Week 38 quiz

Main Objectives

The main objective of this exercise is to be able to describe Threat modeling in the context of mobile applications, REST APIs and secure REST-based communications.

Description of the Assignment

Attempt each of the below questions as they appear and complete them, preferably as classwork (if you get the time) or later as homework 😊

(1) State the benefits of “threat modeling” in a project.

(2) Consider the scenario shown in the below figure. The system consists of health data (heart rate and breath rate) monitoring smartwatch that communicates with a mobile phone using Bluetooth. The mobile phone is also connected to a cloud-based health information storage (i.e., stores the readings pushed by the smartwatch) via an App. The App uses a web interface, and the App does not know or care if the connection uses WiFi or a cellular data link. With respect to the given scenario, identify a threat for each of the threat categories in STRIDE (spoofing, tampering, repudiation, information disclosure, denial of service, elevation of privilege) and propose suitable countermeasures to address each of the identified threats.

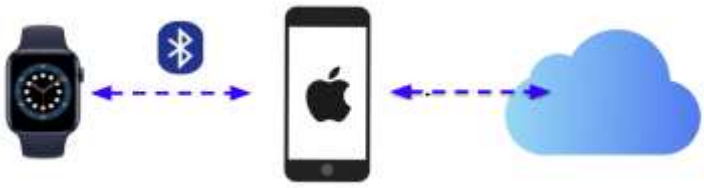


Figure: Threat modeling scenario.

(3) What is a REST interface?

(4) List the features of REST-based Web services.

(5) Describe a Web resource in the context of the REST interface.

(6) Explain how the OAuth 2.0 works?

(7) What is the Internet of Things? Explain the role of mobile applications in IoT.